

Alliances associated

Wi-Fi: Wi-Fi Alliance. Wi-Fi is associated with Wi-Fi Alliance that certifies it in both 5GHz and 2.4GHz range for high performance and better data-transfer rates. After bringing in the IEEE 802.11ac standard, Wi-Fi Alliance is further planning to introduce two new specifications very soon. One will be called WiGig that will run at 60GHz and will be capable to transfer data at tremendous speeds. But, it will have a low range and will not be able to penetrate walls. Whereas, the other one will be called IEEE 802.11ah that will run at 900MHz and cover long ranges at very low speeds.

Bluetooth Smart: Bluetooth SIG. Not connected to any particular alliance, Bluetooth Smart is a technology adopted by Bluetooth Special Interest Group (SIG). Alliance for Wireless Power (A4WP) has already integrated Bluetooth Smart with A4WP standards for smart wireless charging stations, allowing quick charging with improved power control features. Although SIG has considered to improve speed and wireless range for future specifications, there is no information on the new specification. Do you think it would be called Bluetooth4.2?

ZigBee: ZigBee Alliance. It is an open association that provides low power, green and global standards for wireless networking. Teams are working on improving and extending ZigBee standards. Although the Alliance does not intend to come up with a new ZigBee specification, they are working on adding new functionality using RF4CE specification.

WiMax: WiMax Forum. Although not an Alliance, WiMax Forum created WiMax. Currently, it takes care of expanding this technology worldwide, enhancing its features by overcoming challenges and bringing up new standards. In 2009, an Open Patent Alliance was associated with this standard. However, it is not as active as the WiMax Forum. The Forum has not yet disclosed the new standards that will be associated with the technology but is working continuously on effective deployment of WiMax Advance.



Advanced use of Bluetooth

"If the hardware is to be implanted within a human body, it should have stacked data for easy communication."

Less-complex, multicast communication

Previously working with IEEE 802.16m and IEEE 802.16.1 standards, Wireless Metropolitan Area Networks (WiMax) has moved on to high-reliability networks with 802.16n and 802.16.1a standards. Where 802.16n focuses on minimising the complexity and architecture of the systems, 802.16.1a aims to promote commercialisation of broadband wireless access systems.

WiMax gained importance amongst consumers with the advent of WiMax Advance. WiMax Advance had certain host challenges and the future of WiMax technology was uncertain, so the WiMax Forum came up with the WiMax 2.1 initiative. This initiative provides a clear technical roadmap for WiMax Advance and its implementation with Time Domain Long Term Evolution (TD-LTE). Also, the Forum identified the need for multiple broadband wireless access technologies within the WiMax Advance network. The implementation of WiMax Advance with TD-LTE was done for the interoperability of products.

Such standards, when associated with WiMax, take it to a new level, providing minimum architectural complexity and avoiding any excessive system complexity. The interoperability of the IEEE 802.16n standard allows you to access multiple networks. It also provides multicast communication with multicast key management (MKM) protocols to avoid any eavesdropping and enable a reliable network.

The interface involving the devices between high reliability (HR) stations

Other features of WiMax Advance

- Minimises architectural complexity and avoids excessive system complexity.
- Supports low-cost devices and minimises the total ownership cost.
- Provides a high degree of assurance for maintaining sufficient connectivity as compared to previous standards.
- Provides specifications to develop wireless access systems for public protection and disaster relief (PPDR).
- Promotes the commercialisation of broadband wireless access systems amongst manufacturers.

Some useful links

To know more about the wireless communication technology, please refer to the following articles published at

- <http://electronicsforu.com>
- <http://tinyurl.com/kfvdv3> – Wireless Technologies with their alternatives
- <http://tinyurl.com/osdc9jh> – Wireless Products
- <http://tinyurl.com/q2fxcy2> – Plasma Antenna
- <http://tinyurl.com/nygkxif> – Li-Fi

Increasing challenges for the designers

Over time, all these technologies have evolved to such an extent that designers find it difficult to make suitable design decisions. Pranav Vishwanathan, business manager, TETCOS explains, "Consider the design and fabrication of a new wireless technology. During the design phase, the designer is presented with a myriad of decisions and it would be very costly to actually fabricate all of the potential designs as a means of evaluating their respective performance."

In reality, the requirement is to:

- Investigate the relative superiority of each design.
- Determine the correctness and efficiency of a design before the system is actually constructed.
- Investigate the effects of specific design decisions during the design phase rather than the construction phase.
- Study a problem at several different levels of abstraction
- Understand the behaviours and interactions of all the high-level components within the system

2.4GHz band over three channels, it gets the power-saving mechanism incorporated. It utilises the industry standard AES-128 security scheme that makes the technology more secure. The RF4CE specification with Remote Control 2.0 standard allows ZigBee modules to control a wide range of products, starting from the home entertainment system to the key-less entry system.

Replacing IR with RF in ZigBee

initiates some design challenges when it comes to hardware and standards. Satyam Raj, design engineer, Toradex says, "Systems break with regards to RF circuits because of electromagnetic compatibility (EMC) and the hardware needs to follow certain compliances too." A designer also needs to find out the intent of use of the hardware with RF standards when it comes to RF based wireless networks. Raj explains,